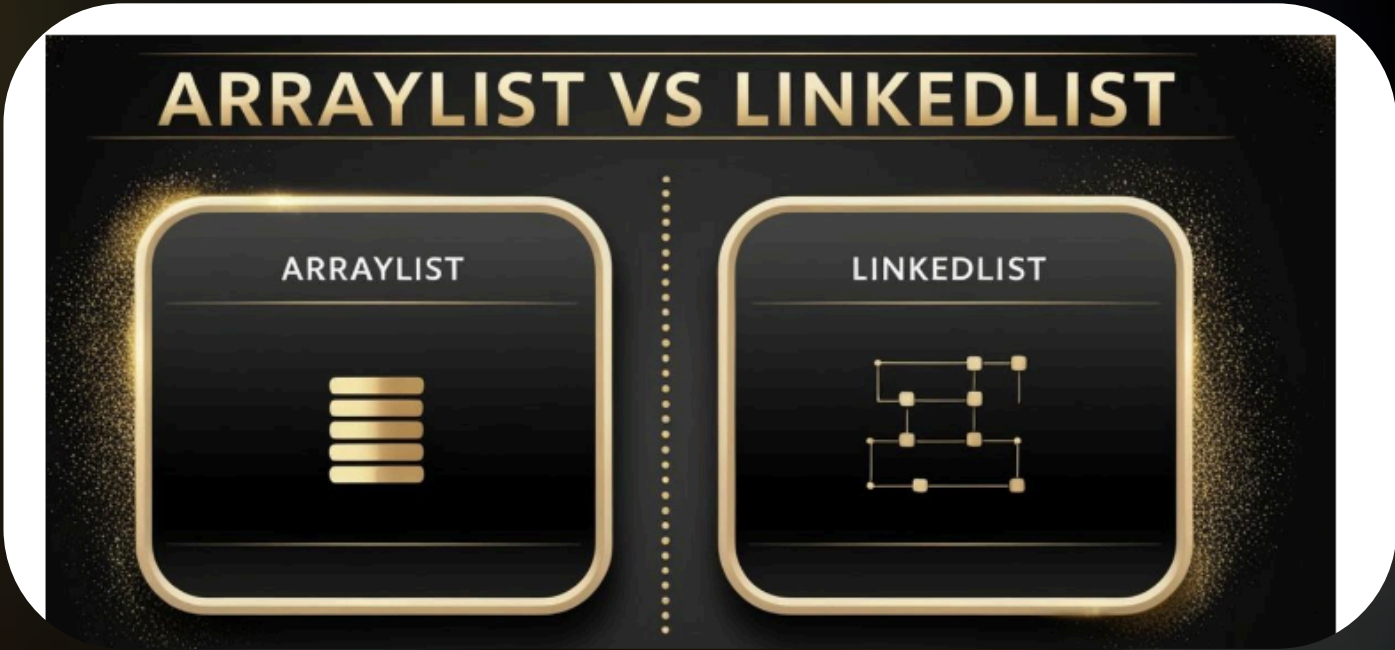




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TOP JAVA COLLECTIONS DIFFERENCES BASED INTERVIEW QUESTIONS



Similarities

- Implements List interface
- Maintains the order of insertion until it disturbs
- Stores the element based on index only
- Non-synchronized classes

Differences between ArrayList and LinkedList

Features	ArrayList	LinkedList
Data structures	▪ It uses dynamic array or resizable array store the elements	▪ It uses doubly linked list DS to store the elements
Act as	▪ ArrayList class can act as a list only ▪ Because it implements List only	▪ LinkedList class can act as a list and queue ▪ Because it implements List and Dequeue interfaces
Suitable for	▪ Suitable for search operation ▪ Provides constant time for search operation ▪ If our application has more searching operation ArrayList is preferable over LinkedList	▪ Suitable for Manipulation ▪ Provides constant time for add and remove operations. ▪ If our application has more for manipulation operations LinkedList is preferable over ArrayList.
Time complexity to Access the element.	▪ O(1) ▪ Via the get and set methods.	▪ O(n/2) ▪ Time complexity to access the elements.
Type: reason	▪ Sequential list ▪ Stores the elements one after another	▪ Non sequential list ▪ Stores the elements randomly

When to use ArrayList and Linked List

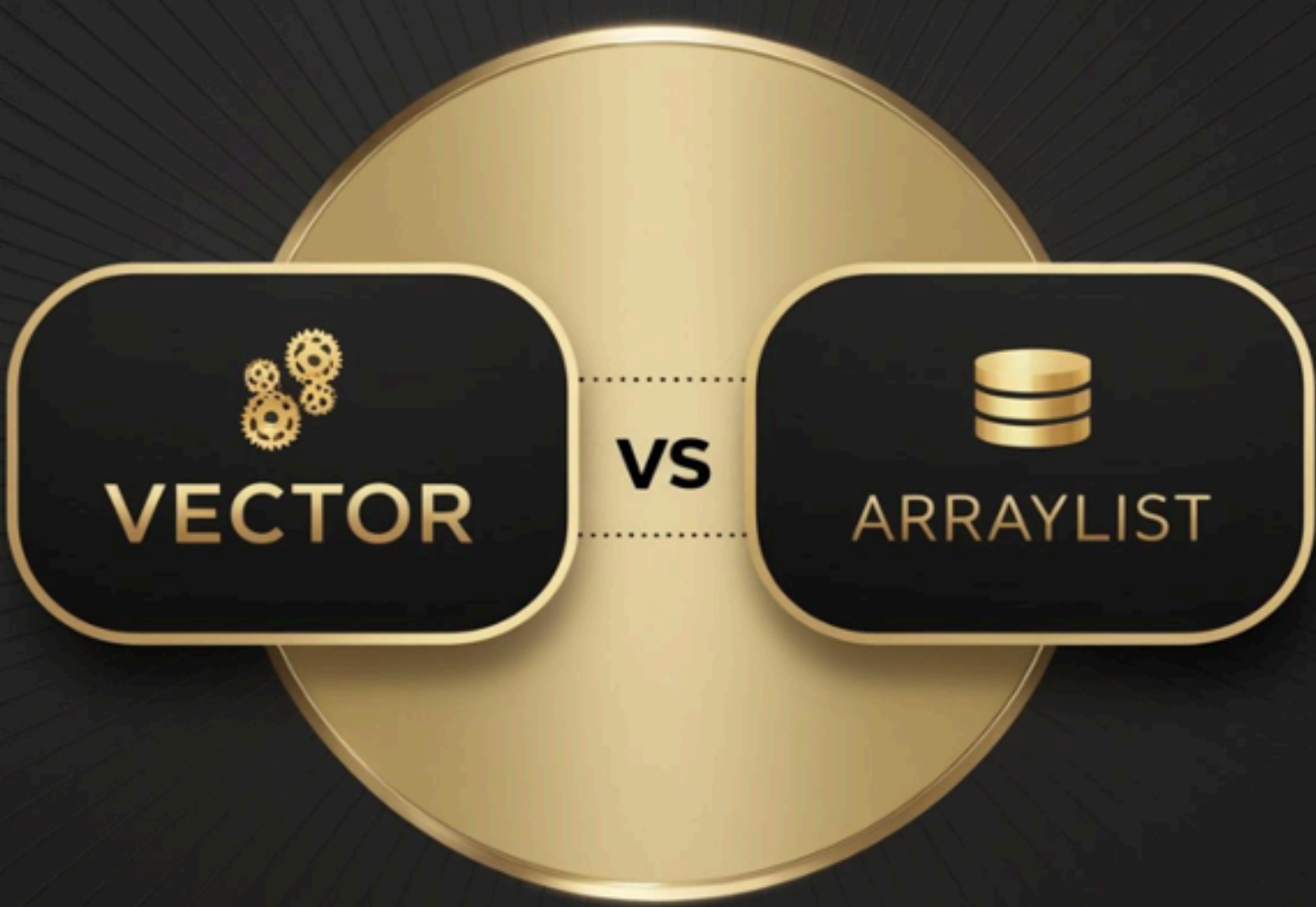
ArrayList	LinkedList
▪ Suitable for search operation ▪ Provides constant time for search operation ▪ If our application has more searching operation ArrayList is preferable over LinkedList	▪ Suitable for Manipulation ▪ Provides constant time for add and remove operations. ▪ If our application has more for manipulation operations LinkedList is preferable over ArrayList.
▪ ArrayList is better to access data	▪ LinkedList is better to manipulate data
▪ Time complexity to Access the element using ArrayList is O(1)	▪ Time complexity to Access the element using LinkedList is O(n/2)



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Features	Vector	ArrayList
Synchronization	It is a Synchronized class	It is not Synchronized class
Thread Safe	Vector is Thread safe Because methods of vector class synchronized.	- ArrayList is not thread safe. - Because methods of ArrayList class are not synchronized.
Traverse elements	- Vector uses Enumeration interface to traverse the elements. - But it can use Iterator also.	ArrayList uses Iterator interface to traverse the elements.
If Current Size Exceeds	- Vector increments 100%. - Means doubles the current size if total number of elements exceeds than its capacity.	if current size exceeds from its capacity Dynamically grows 50% of current array size.
Fast / Slow	Vector is slow because it is synchronized.	ArrayList is fast because it is non synchronized.
Legacy	Vector is a legacy class.	ArrayList is not a legacy class, it is introduced in JDK 1.2.



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HASHMAP VS HASHTABLE



LinkedHashMap

- Subclass of HashMap
- All properties are same as HashMap
- Except it is preserve the order of insertion

HashTable

- It is thread safe class
- Because all the methods of HashTable is synchronized
- HashTable is also called as synchronized Map
- Key doesn't allow NULL value
- Value doesn't allow NULL value

TreeMap

- TreeMap implements sorted map
- Hence elements of TreeMap sorted in Ascending order

HashMap v/s HashTable

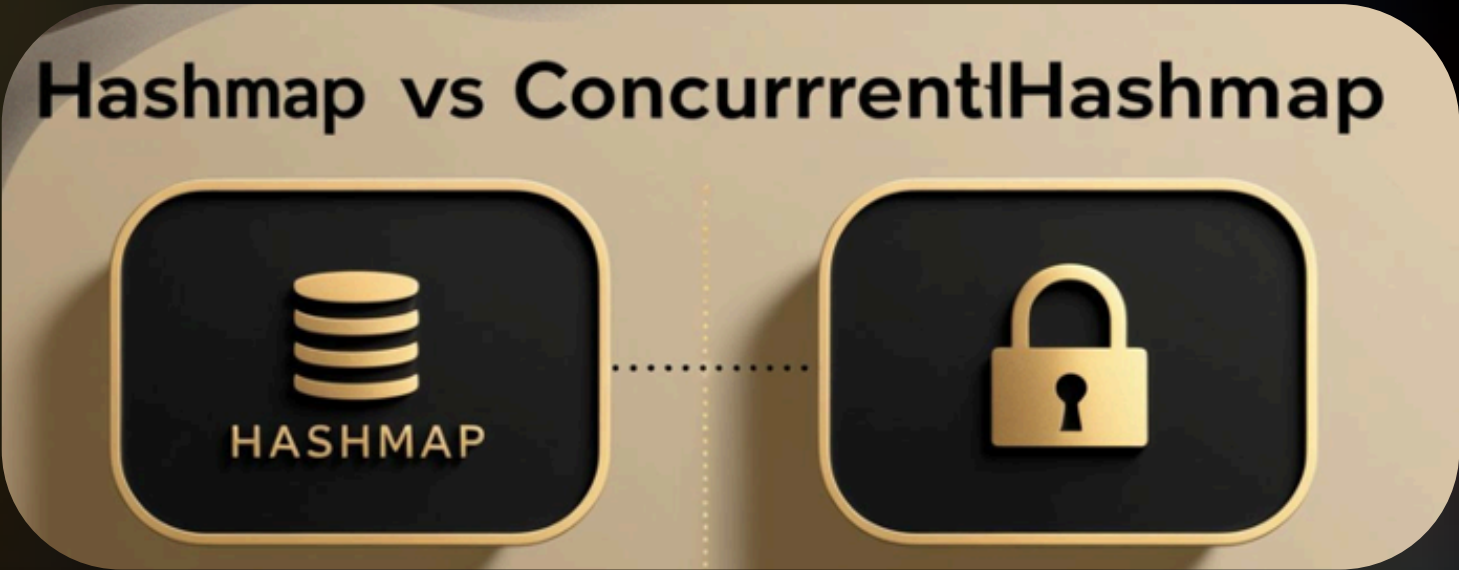
HashMap	HashTable
▪ Non- Synchronized Class	▪ Synchronized Class ▪ Because all the methods of HashTable is synchronized
▪ Not Thread Safe	▪ Thread Safe class
▪ NULL key and values allowed	▪ NULL key and values are not allowed
▪ Not legacy class [JDK 1.2]	▪ Legacy class [JDK 1.0]



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Differences	HashMap 📁	ConcurrentHashMap 🛡️
Thread Safe	❌ Not Thread Safe	✅ Thread Safe Class
Synchronized	🔄 Not synchronized 🔑 Must use SynchronizedMap 🗄️ Locks the entire map when synchronized	🔄 Synchronizes only a portion of the map, ensuring better performance
Null Keys	🔑 Allows one null key	🚫 Does not allow null keys ? Throws NullPointerException
Performance	⚡ Faster than SynchronizedMap and ConcurrentHashMap in single-threaded scenarios 🧑‍🤝‍🧑 Multiple threads can access the code simultaneously	⚙️ Locks only part of the map, reducing contention but slightly impacting performance
ConcurrentModificationException	❌ Throws ConcurrentModificationException when a thread modifies the map while another thread iterates	✅ No ConcurrentModificationException thrown
Iterator	🚫 Fail-Fast	✅ Fail-Safe

Normal	Concurrent
▪ Not Thread safe	▪ Thread safe
▪ Not Synchronized	▪ Synchronized
▪ Not Suitable for Multithreaded env ▪ More waiting time	▪ Suitable for Multithreaded env ▪ No or less
▪ Object level lock	▪ Segment level lock
▪ No concurrency	▪ We can achieve concurrency
▪ Iterator: Fail fast	▪ Fail Safe
▪ ConcurrentModificationException	▪ No
▪ Allows NULL keys but only one	▪ Not allows null : NullPointer
▪ Less Performance in Multithreaded env	▪ More Performance in Multithreaded env



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PURPOSE OF THIS DOCUMENT

- **Use this PDF as a quick last-minute revision guide only.**
- **For more details, please perform R&D**
- **Once the interview series is complete, we will cover this topic in detail.**



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